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The document outlines methods for extracting silver, highlighting mining techniques such as open-pit, underground, and byproduct mining. It detail...

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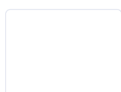
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Extraction of S

Methods, Processes, Applications

Introduction

- Silver is a precious metal with industrial and ornamental use. Extraction involves mining, an refining processes to achieve high purity.

Mining Methods

- 1. "Open-Pit Mining": Surfsilver ores like argentite.
- 2. "Underground Mining": ore deposits from beneath the surface.
- 3. "Byproduct Mining": Silver is obtained as a byproduct of gold, copper, or

AD

Extraction Process

- 1. "Roasting": Used to convert silver ore from other minerals.
- 2. "Leaching": Silver is dissolved in cyanide solution:
$$\text{Ag}_2\text{S} + 4\text{NaCN} \rightarrow 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$$
- 3. "Precipitation": Silver is precipitated from solution using zinc:
$$2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \rightarrow 2\text{Na}[\text{Zn}(\text{CN})_4]$$

Smelting and Refining

- 1. "Cupellation": Heating silver removes impurities.
- 2. "Electrolytic Refining": Produces pure silver.
- 3. "Miller Process": Silver is refined using chlorine gas to remove contaminants.

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Environmental Impact

- 1. Cyanide use in leaching can harm the environment.
- 2. Land degradation and mining waste.
- 3. Efforts to reduce impacts:
 - Recycling silver from used products.
 - Using eco-friendly leaching.

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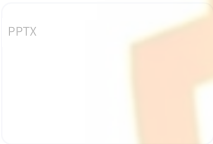
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1. **Extraction of Silver Methods**, Processes, and Applications
2. **Introduction • Silver** is a precious metal with numerous industrial and ornamental applications. Its extraction involves mining, smelting, and refining processes to achieve high purity.
3. **Mining Methods • 1.** ****Open-Pit Mining****: Surface mining for silver ores like argentite. • 2. ****Underground Mining****: Extracting silver ore deposits from beneath the surface. • 3. ****Byproduct Mining****: Silver is obtained as a byproduct of gold, copper, and lead mining.
4. **Extraction Processes • 1.** ****Froth Flotation****: Used to separate silver ore from other minerals. • 2. ****Leaching****: Silver is dissolved using cyanide solution: • $\text{Ag}_2\text{S} + 4\text{NaCN} \rightarrow 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$. • 3. ****Precipitation****: Silver is recovered from solution using zinc. • $2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \rightarrow 2\text{Ag} + \text{Na}_2[\text{Zn}(\text{CN})_4]$.
5. **Smelting and Refining • 1.** ****Cupellation****: Heating silver with lead removes impurities. • 2. ****Electrolytic Refining****: Electrolysis produces pure silver. • 3. ****Miller Process****: Silver is refined using chlorine gas to remove contaminants.
6. **Environmental Impacts • 1.** Cyanide use in leaching can harm the environment. • 2. Land degradation and mining waste. • 3. Efforts to reduce impacts: • - Recycling silver from used products. • - Using eco-friendly leaching alternatives.
7. **Applications of Silver • 1.** Jewelry and tableware. • 2. Electronics: Conductors and contacts. • 3. Photography: Used in film production. • 4. Medicine: Antimicrobial and wound healing. • 5. Renewable Energy: Silver paste in solar cells.
8. **Conclusion • The extraction** of silver is a multi-step process involving mining, chemical treatments, and refining. Advances in technology and sustainable practices are essential for meeting the growing demand for silver while minimizing environmental impact.

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